

1. Administrative & Study Identification

Full Study Title: A Phase II, Multicenter, Randomized, Placebo-Controlled, Double-Blind Study to Evaluate the Pharmacodynamics, Safety, Tolerability, Pharmacokinetics, and Efficacy of R07204239 in Participants With Facioscapulohumeral Muscular Dystrophy **Short Title/Acronym:** MANOEUVRE Study **Protocol Number:** BN43703 **NCT Number:** NCT05548556 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** R07204239 (GYM329 / emugrobarb) - a monoclonal anti-myostatin antibody **Phase of Development:** Phase II **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the percentage change from baseline in the contractile muscle volume of the quadriceps femoris muscle as assessed by magnetic resonance imaging (MRI) at Week 52, alongside the safety profile of R07204239. **Secondary Objectives:** To evaluate the change from baseline of muscle volume in various muscles and muscle groups as assessed by MRI; to assess motor function and strength; and to evaluate the pharmacodynamics (PD), tolerability, and pharmacokinetics (PK) of R07204239.

2.2 Background & Rationale Facioscapulohumeral muscular dystrophy (FSHD) is a rare genetic muscle disorder characterized by progressive asymmetric skeletal muscle weakness and atrophy. The condition worsens slowly over time and can lead to severe mobility issues. Currently, there are no approved disease-modifying drugs for FSHD, representing a high unmet medical need. R07204239 binds to human latent myostatin to block its activity, hypothesizing that suppressing myostatin could safely promote an increase in muscle size and strength.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1 (R07204239 vs. Placebo)

3.2 Planned Enrollment & Duration Target N\$: ~48 to 51 participants **Number of Sites:** Multicenter (Global) **Estimated Study Start:** September 2022 **Estimated Primary Completion:** Q1 2026 (Note: Clinical development in this indication was halted in early 2026 following a planned data review)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult ambulant patients (able to walk independently). 2. Genetically confirmed diagnosis of FSHD1 or FSHD2.

4.2 Exclusion Criteria 1. Pregnancy or breastfeeding, or intention of becoming pregnant during the study or within 17 months after the final dose. 2. Current or previous treatment (or receipt) of anti-myostatin therapies. 3. Treatment with any investigational therapy within 90 days prior to screening, or 5 drug-elimination half-lives. 4. Contraindications to MRI scans. 5. Presence of clinically significant ECG abnormalities or cardiovascular disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered every 4 weeks. **Route of Administration:** Subcutaneous (SC) injection in the abdomen. **Duration of Treatment:** 52 weeks of double-blind treatment (placebo or active), followed by an optional 52-week Active Treatment Extension phase (all participants receive R07204239). Total maximum treatment is 104 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for FSHD (e.g., physical therapy, pain management). **Prohibited:** Concurrent anti-myostatin therapies or other investigational drugs.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Week -4 to 0	Informed Consent, genetic confirmation, baseline movement data collection via wearable device
Baseline	Day 0	Randomization, Baseline MRI, First Dose of R07204239 or Placebo
Interim	Every 4 Weeks	Treatment administration, Safety Checks, Efficacy Assessments (motor function)
End of Study	Week 52 / Week 104	Final MRI at W52 (Primary Endpoint), extension phase completion (W104), followed by a 6-month safety follow-up period

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage change from baseline in the contractile muscle volume of the quadriceps femoris muscle at Week 52. **Measurement Method:** Assessed by Magnetic Resonance Imaging (MRI).

7.2 Secondary & Exploratory Endpoints * Change from baseline of muscle volume in other various muscles and groups of muscles (by MRI). * Assessment of motor function and strength. *

Pharmacokinetics (PK) and Pharmacodynamics (PD) profiling of R07204239.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Conducted continuously at hospital visits scheduled at least every 4 weeks to monitor treatment response and potential side effects. **Serious Adverse Events (SAEs):** Handled via standard 24-hour reporting protocols. (Note: The drug was well-tolerated, and no SAEs led to trial withdrawals). **Data Monitoring Committee (DMC):** Independent safety monitoring board maintained oversight.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy; Safety Set for AE evaluations. **Sample Size Justification:** A target of ~48-51 participants was designed to provide adequate statistical power to detect meaningful changes in MRI-based muscle volume over 52 weeks in a Phase II setting. **Handling of Missing Data:** Standard mixed-model repeated measures (MMRM) applied for continuous efficacy endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard onsite monitoring alongside remote review of wearable device movement data. **Audit Trail:** Robust centralized MRI reading and standardized eCRF locking mechanisms utilized to ensure data integrity.

11. Study Identifiers & Governance

Primary ID: BN43703 **Registry Number:** NCT05548556 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** MANOEUVRE Study **Official Regulatory Title:** A Phase II, Multicenter, Randomized, Placebo-Controlled, Double-Blind Study to Evaluate the Pharmacodynamics, Safety, Tolerability, Pharmacokinetics, and Efficacy of R07204239 in Participants With Facioscapulohumeral Muscular Dystrophy

12. Clinical Framework (FSHD Specific)

Primary Condition: Facioscapulohumeral Muscular Dystrophy (FSHD) **Diagnosis Threshold:** Genetically confirmed FSHD1 or FSHD2 diagnosis in ambulant adults **Medical Methodology:** Experimental targeted Anti-myostatin Therapy **Optimization Target:** Increase in contractile muscle volume and motor function preservation

13. Study Procedures & Methodology

Management Pathway: Subcutaneous delivery of targeted monoclonal antibodies **Education Protocol (HCP):** Site training for proper SC injection and MRI standardization protocols **Education Protocol (Patient):** Training on compliance with wearable device utilization during the pre-treatment and follow-up periods **Quality Audit Mechanism:** Centralized imaging core lab review and wearable data compliance tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to 104 treatment weeks + 24 weeks (6 months) safety follow-up (total ~2.5 years) **Onsite Visit Cadence:** Every 4 weeks for drug administration and evaluation **Remote Monitoring Frequency:** Continuous monitoring via wearable device (initiated in the 4-week pre-treatment phase) **Checklist Review Interval:** Routine continuous assessment

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					